

Week 14

Mechanical Noise Control

Spring 2021
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Noise control techniques

- Rooftop AHU
- Rumble noise control
- Breakout noise control
- Plenums
- Duct attenuation – Lined vs unlined
- Elbows
- Silencers

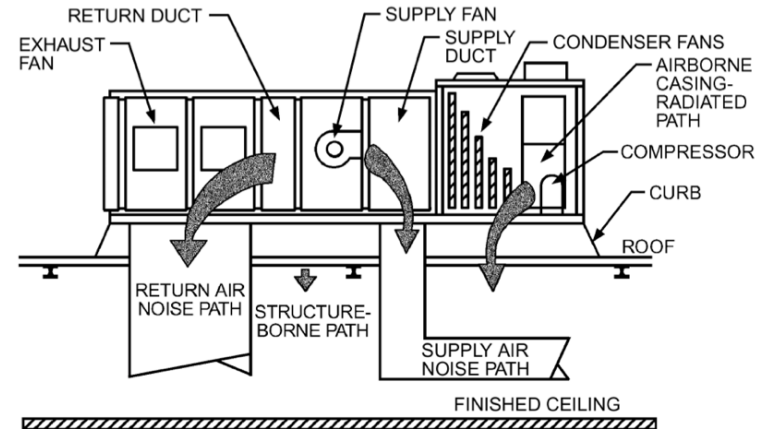
FANS

- Fans selected for peak efficiency
 - with variable airflow systems:
 - *insure efficiency & stability across operating range*
- Select quiet equipment
 - per AMCA 300 method – bare fan L_w data
 - per AHRI 260 method – unit level L_w data
 - Includes filters, coils, plenums, casing, etc.
- Good flow conditions

ROOFTOP AHU's

Rooftop air handling units

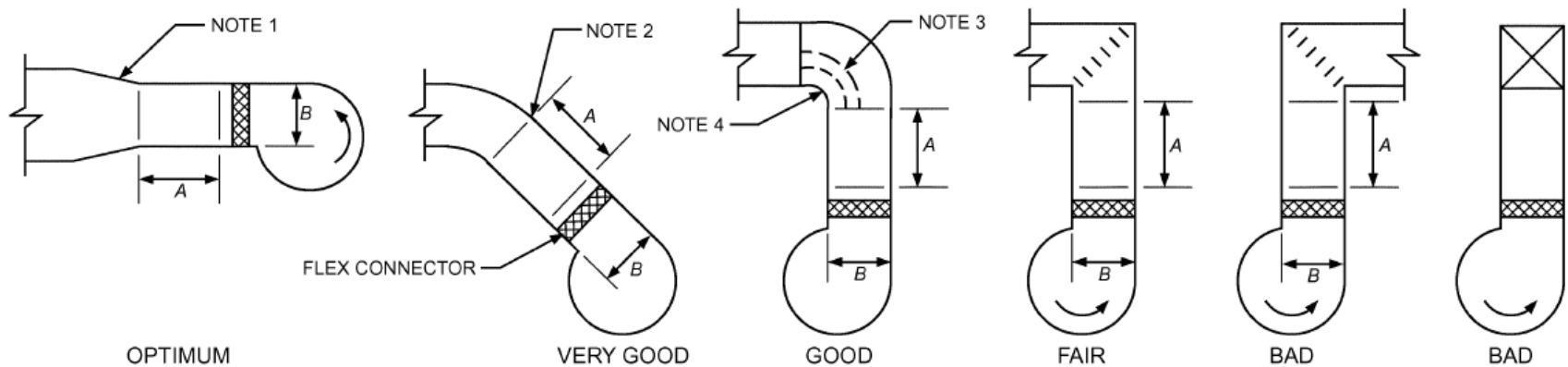
- Locate away from noise-sensitive spaces and keep from over windows and lightweight partitions
- Treat ductborne and breakout noise paths
- Treat radiated path (outside air, exhaust, inside curb)
- Adequately isolate vibration
- Limit roof penetrations to size of ductwork and piping, seal airtight
- Run ducts over roof for several duct diameters before entering building
- Raise on structural steel supported from columns, if possible



RUMBLE

FAN DISCHARGE CONDITIONS – *MINIMIZE RUMBLE*

- Use round duct which is considerably stiffer than rectangular ductwork
- Lag ductwork with drywall (as shown for breakout noise in later slide)
- Construct enclosure around ductwork (as shown in later slide)
- Minimize turbulent flow conditions immediately at fan discharge
- Provide straight lengths without abrupt changes in flow direction or size transition
- Use flexible connector to minimize vibration transmission along metal ductwork
- Turning vanes help minimize turbulence through elbows



Notes:

1. Slopes of 1 in 7 preferred. Slopes of 1 in 4 permitted below 2000 fpm.
2. Dimension A should be at least 1.5 times B, where B is largest discharge duct dimension.
3. Rugged turning vanes should extend full radius of elbow.
4. Minimum 6 in. radius required.

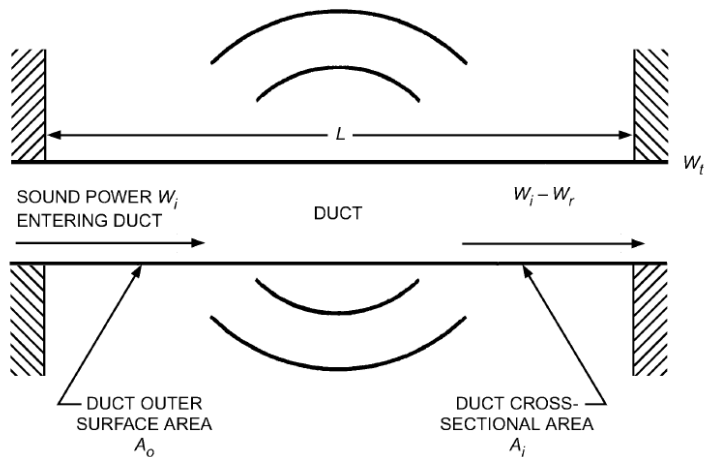
Various outlet configurations for centrifugal fans and their possible rumble conditions

BREAKOUT NOISE

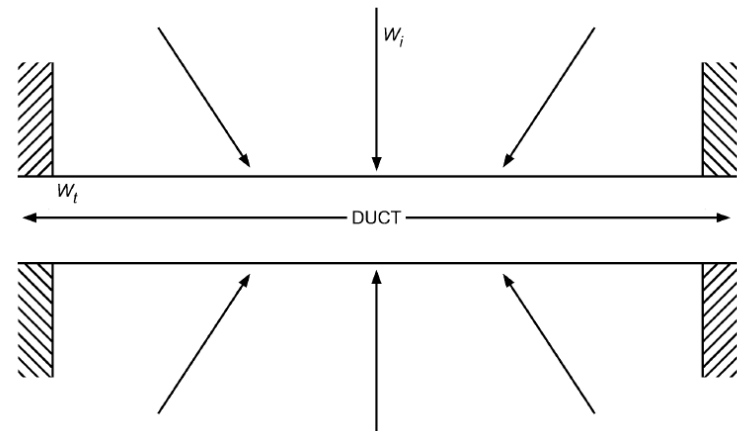
BREAKOUT (DUCT RADIATED) AND BREAK-IN NOISE

- Noise within ductwork can radiate out from the surfaces of the duct into the surrounding space
- Breakout noise from ductwork above sensitive areas requires treatment
- Ductwork closest to the AHU is often problematic due to high sound power generated from fan
- Break-in noise is a condition where loud noise from the space penetrates the ductwork which then readily carries the noise to other spaces/rooms

Breakout Noise



Break-In Noise

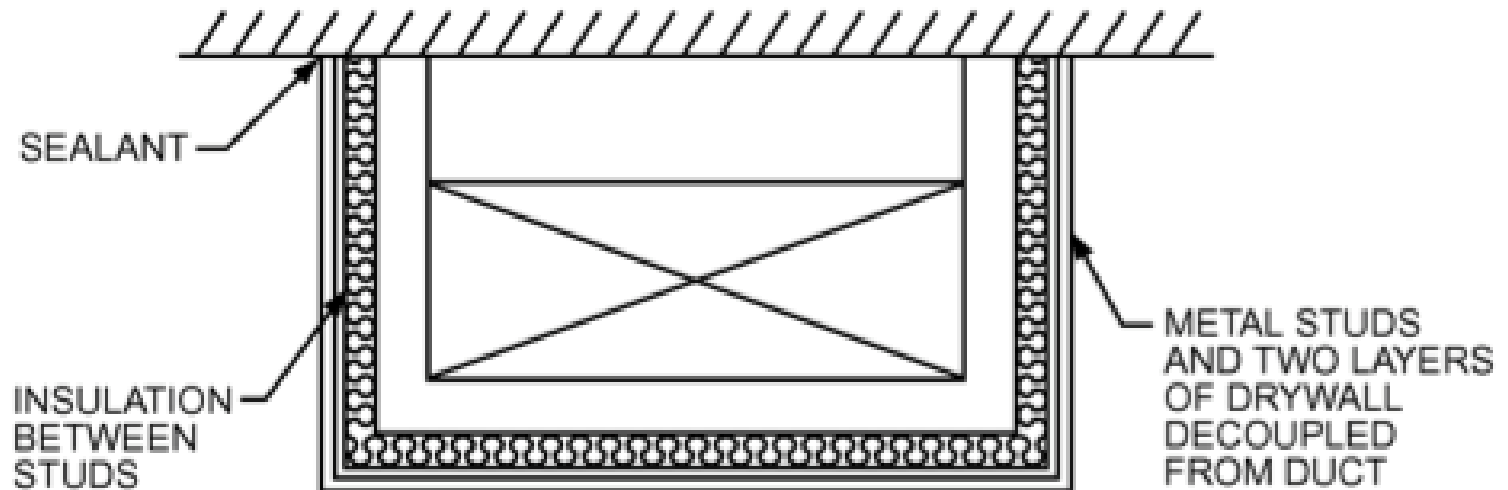


BREAKOUT NOISE

- Is a function of duct TL, surface area of duct, and liner (i.e., absorption) within duct
- Can be useful to release sound power into non-critical areas (e.g., outdoors, mechanical rooms, etc.)
 - Example: For rooftop AHU, run ducts over roof for several duct diameters before entering building to reduce sound power in ductwork
- To limit breakout noise:
 - Round metal ducts are stiffer and less susceptible to breakout noise at low frequencies versus rectangular ducts
 - Add absorption within duct (i.e., duct liner), where thicker is better
 - Use heavy gauge sheet metal and/or lag the duct with heavy construction material, such as one to two layers of drywall, which will also help with rumble

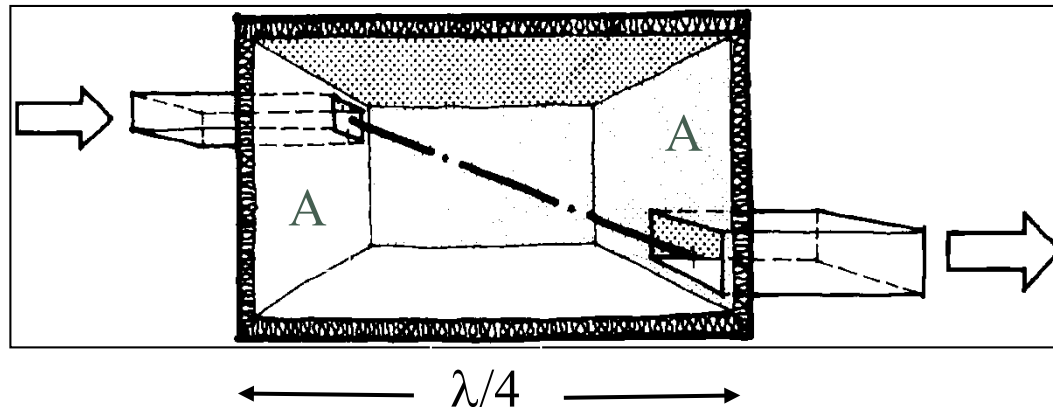
ACOUSTIC ENCLOSURES

- An acoustical enclosures provide better transmission loss than lagging
 - *Enclosure is decoupled from duct – vibration break*
 - Air gap between “two-leaf partition” is larger, so better TL
- Effective for duct rumble (16-100 Hz) if air gap is 6” or more



Acoustical Plenums

- A plenum is a large box or space with an inlet and one or more outlets.
- Very effective attenuation (especially for fan noise) if space and pressure drop are available
 - Often placed between fan and main air distribution ducts to smooth turbulence
 - Typically has a 1 to 6 inch thick fiberglass liner for low freq. α
 - Total lined surface area should be at least 10 times greater than the inlet area
 - Both end wall areas 'A' should be 4 to 6 times the areas of ducts penetrating the end walls
 - λ is lowest wavelength attenuated by the plenum, as shown



SELF-GENERATED NOISE

Aerodynamically generated sound in ducts

- Follow airflow velocity recommendations
- Minimize velocity pressure drop – avoid successive fittings

Table 8 Maximum Recommended Duct Airflow Velocities to Achieve Specified Acoustic Design Criteria

Main Duct Location	Design RC(N)	Maximum Airflow Velocity, fpm	
		Rectangular Duct	Circular Duct
In shaft or above drywall ceiling	45	3500	5000
	35	2500	3500
	25	1700	2500
Above suspended acoustic ceiling	45	2500	4500
	35	1750	3000
	25	1200	2000
Duct located within occupied space	45	2000	3900
	35	1450	2600
	25	950	1700

Notes:

1. Branch ducts should have airflow velocities of about 80% of values listed.
2. Velocities in final runouts to outlets should be 50% of values or less.
3. Elbows and other fittings can increase airflow noise substantially, depending on type. Thus, duct airflow velocities should be reduced accordingly.

Table 9 Maximum Recommended Air Velocities at Neck of Supply Diffusers or Return Registers to Achieve Specified Acoustical Design Criteria

Type of Opening	Design RC(N)	“Free” Opening Airflow Velocity, fpm
Supply air outlet	45	625
	40	560
	35	500
	30	425
	25	350
Return air opening	45	750
	40	675
	35	600
	30	500
	25	425

Note: Table intended for use when no sound data are available for selected grilles or diffusers, or no diffuser or grille is used. The number of diffusers or grilles increases sound levels, depending on proximity to receiver. Allowable outlet or opening airflow velocities should be reduced accordingly in these cases.

Duct attenuation

- Sound is attenuated as it travels down the length of sheet metal duct
- Low frequency attenuation is largely due to breakout noise which is why attenuation is greater for rectangular ducts at 250 Hz and below

Table 16 Sound Attenuation in Unlined Rectangular Sheet Metal Ducts

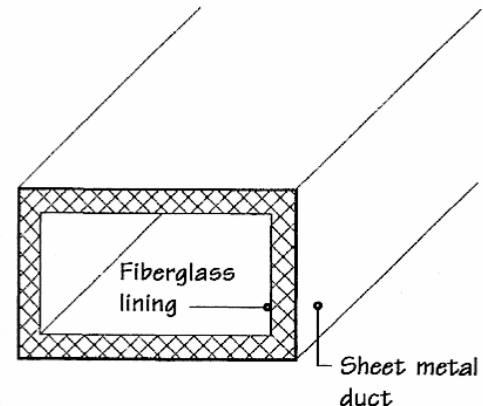
Duct Size, in.	P/A , 1/ft	Attenuation, dB/ft Octave Midband Frequency, Hz			
		63	125	250	>250
6 × 6	8.0	0.30	0.20	0.10	0.10
12 × 12	4.0	0.35	0.20	0.10	0.06
12 × 24	3.0	0.40	0.20	0.10	0.05
24 × 24	2.0	0.25	0.20	0.10	0.03
48 × 48	1.0	0.15	0.10	0.07	0.02
72 × 72	0.7	0.10	0.10	0.05	0.02

Table 19 Sound Attenuation in Unlined Straight Round Ducts

Diameter, in.	Attenuation, dB/ft Octave Midband Frequency, Hz						
	63	125	250	500	1000	2000	4000
$D \leq 7$	0.03	0.03	0.05	0.05	0.10	0.10	0.10
$7 < D \leq 15$	0.03	0.03	0.03	0.05	0.07	0.07	0.07
$15 < D \leq 30$	0.02	0.02	0.02	0.03	0.05	0.05	0.05
$30 < D \leq 60$	0.01	0.01	0.01	0.02	0.02	0.02	0.02

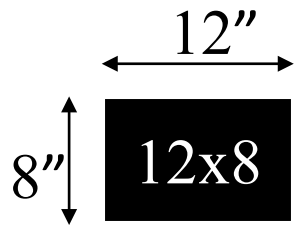
Acoustical Duct Liner

- **Acoustical duct liner** is generally 1 or 2 inch thick semi-rigid fiberglass with a coating **to help prevent flaking of the fiberglass and even microbial growth**
- **Duct liner absorbs sound** as the sound propagates down the duct
- Similar to an acoustical wall panel, the thickness of the absorber affects the absorption coefficient (actually the insertion loss) of the liner
 - **Insertion loss** – Additional sound reduction provided by the fiberglass lining, acoustical plenum or sound attenuator.
 - For ducts, reported in dB of attenuation per length of duct (e.g., dB/ft).

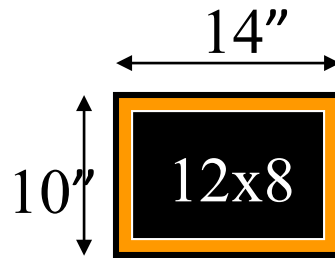


Acoustical Duct Liner

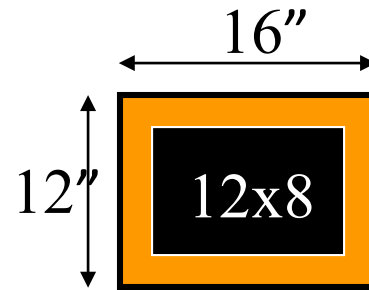
- Clear area is labeled on drawings
 - 1" lining adds 2" to outer duct dimensions
 - 2" lining adds 4" to outer duct dimensions



No lining



1" lining



2" lining

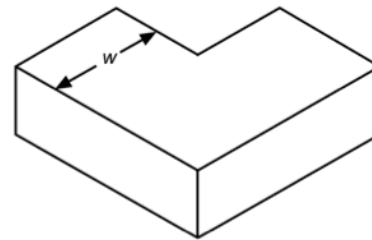
Exterior duct insulation *does not* attenuate ductborne or breakout noise!

ELBOW ATTENUATION

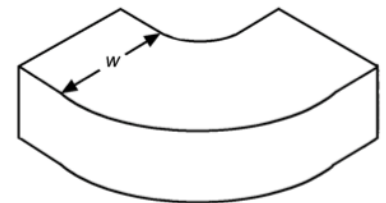
- **Unlined and lined 90° elbows**
 - Greater attenuation:
 - For larger elbows
 - For square elbows (versus radiused)
 - Without turning vanes
 - With acoustic lining



Turning Vanes



SQUARE ELBOW

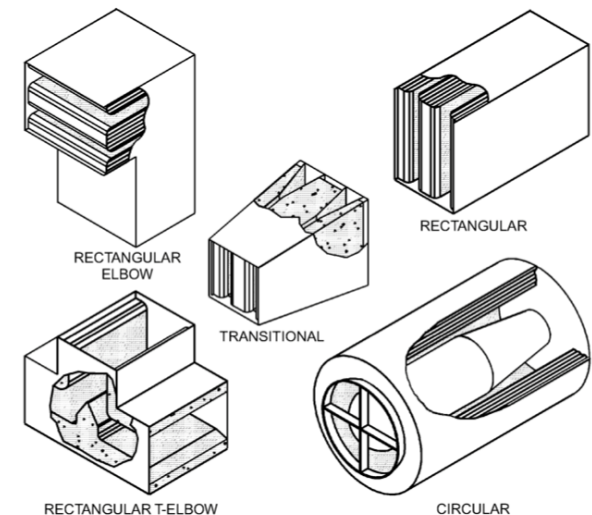
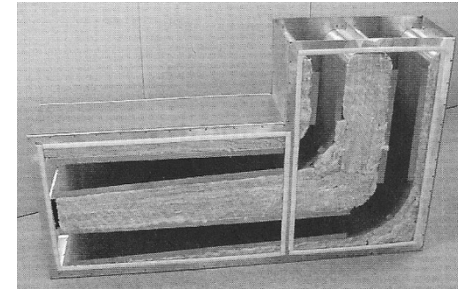


RADIUSED ELBOW

Duct Sound Attenuators

SILENCERS

- Duct sound attenuators, also called **silencers**, sound traps or mufflers
- a length of duct with two or more passages (sometimes one) with thick sound absorbing lining behind a perforated metal facing
- Sound attenuators perform much better at low freq. than duct liner
- Sound travels upstream *and* downstream, so **sound attenuators are most often needed for both paths.**



DUCT SOUND ATTENUATORS

SILENCER TYPES

- **Dissipative** – uses a porous absorber for energy dissipation behind a perforated metal facing
- **Fiber/media free (reactive)** – no absorptive fill, utilizes tuned resonators behind perforated metal
 - IL performance more limited than dissipative
- **Active** – utilizes noise cancellation technology to attenuate primarily low frequency
 - No pressure loss or airflow generated noise

DUCT SOUND ATTENUATORS

SILENCER PERFORMANCE

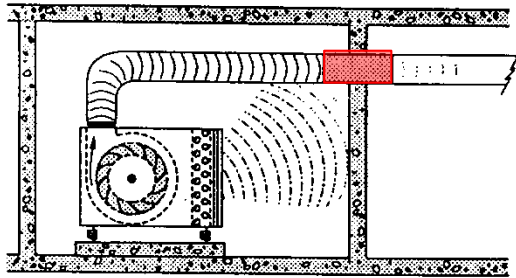
- **Insertion Loss (IL)** – dB of attenuation in each octave band
- **Dynamic Insertion Loss (DIL)** – tested with airflow
 - IL varies depending on direction of airflow
- **Pressure drop (PD)** – pressure loss across attenuator with smooth, developed entry and exit flow
 - Varies with air velocity
 - System pressure drop accounts for losses as installed. This has to be calculated.
- **Self-generated noise** – sound energy created from turbulent flow in attenuator
 - Adds sound back into the system
 - Can be an issue if at the end of a run or for high airflow velocities

DUCT SOUND ATTENUATORS

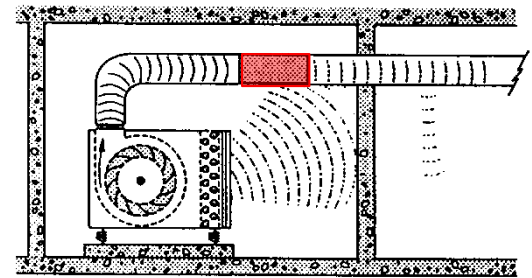
SILENCER STRATEGIC LOCATION

Locate attenuator for minimum breakout (sound escaping through duct wall) & break-in (opposite direction, and a concern if louder than sound in duct, say, on the quiet side of an attenuator)

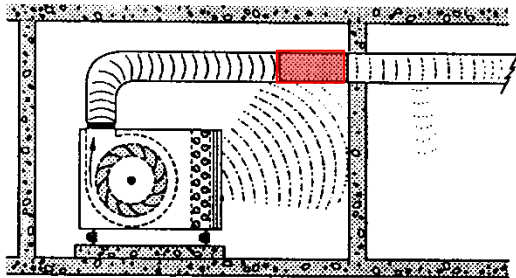
best



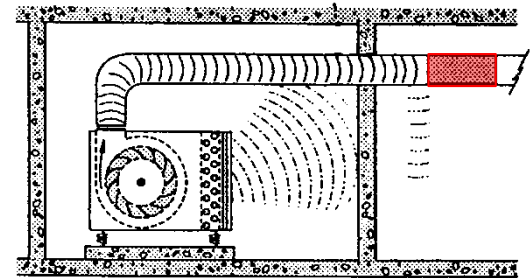
fair



very good

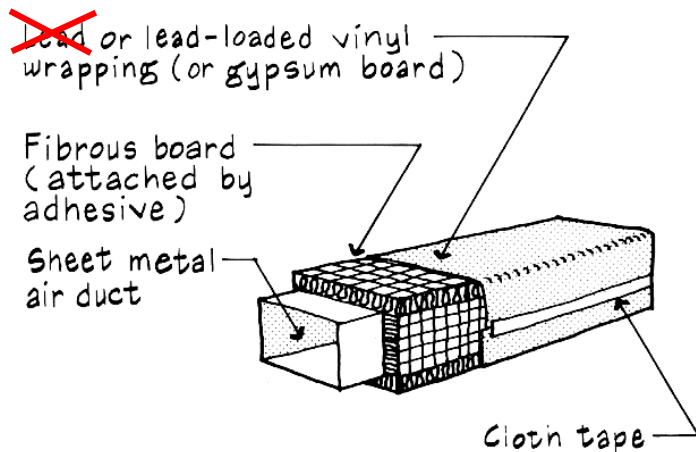


poor

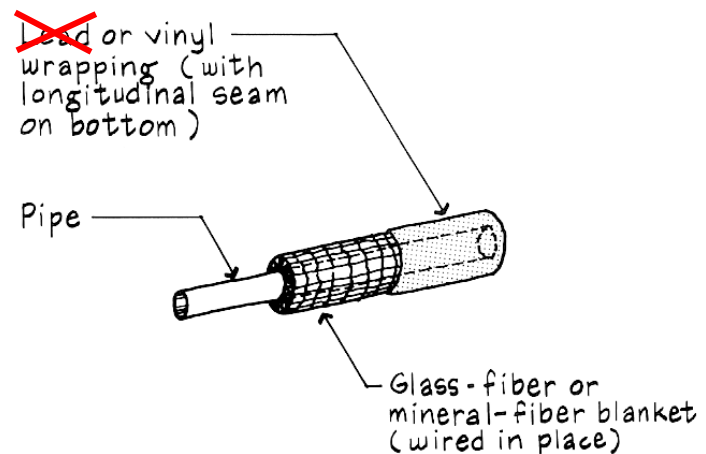


Duct & Pipe Lagging

- **Lagging** is a system of a heavy barrier material wrapped over a porous absorber
- Lagging **reduces breakout & break-in noise** and ring frequency
- Drywall & mass loaded vinyl are common barrier materials.
- Lagging increases outside dimensions, with drywall increasing the dimensions the most.
- Decoupler – fiberglass layer (or other absorber) under lagging provides better performance (like a two-leaf partition)



Duct Lagging Details



Pipe Lagging Details

Air Devices (Diffusers, Grilles, Registers)

Manufacturers rate by NC, but rating is only valid for uniform flow with proper alignment into diffuser

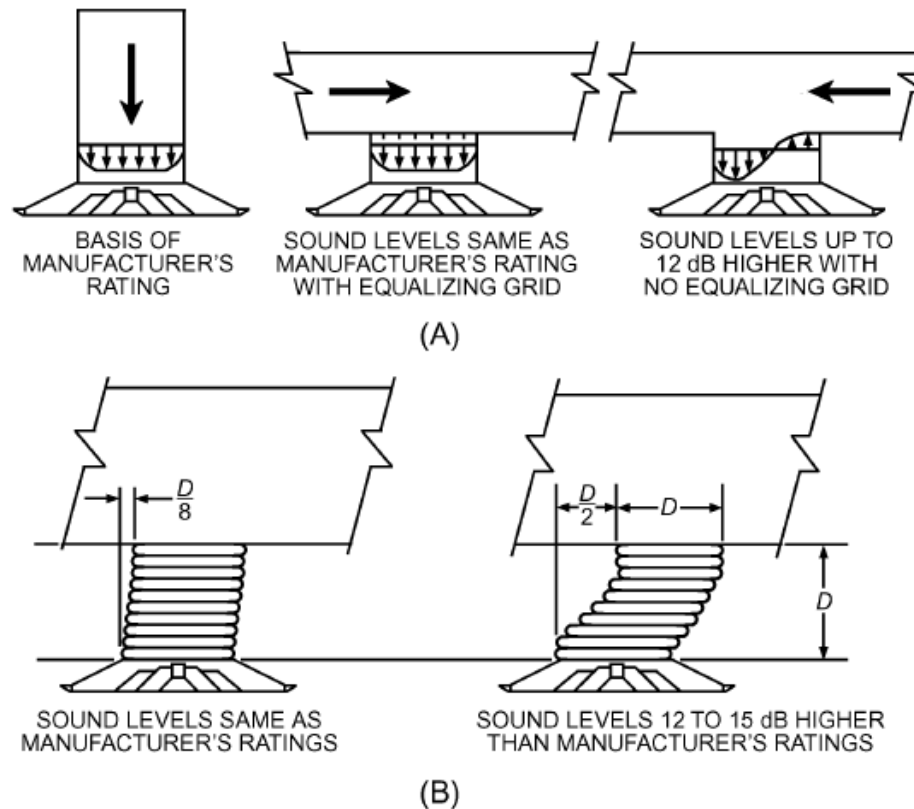


Fig. 14 (A) Proper and Improper Airflow Condition to an Outlet; (B) Effect of Proper and Improper Alignment of Flexible Duct Connector

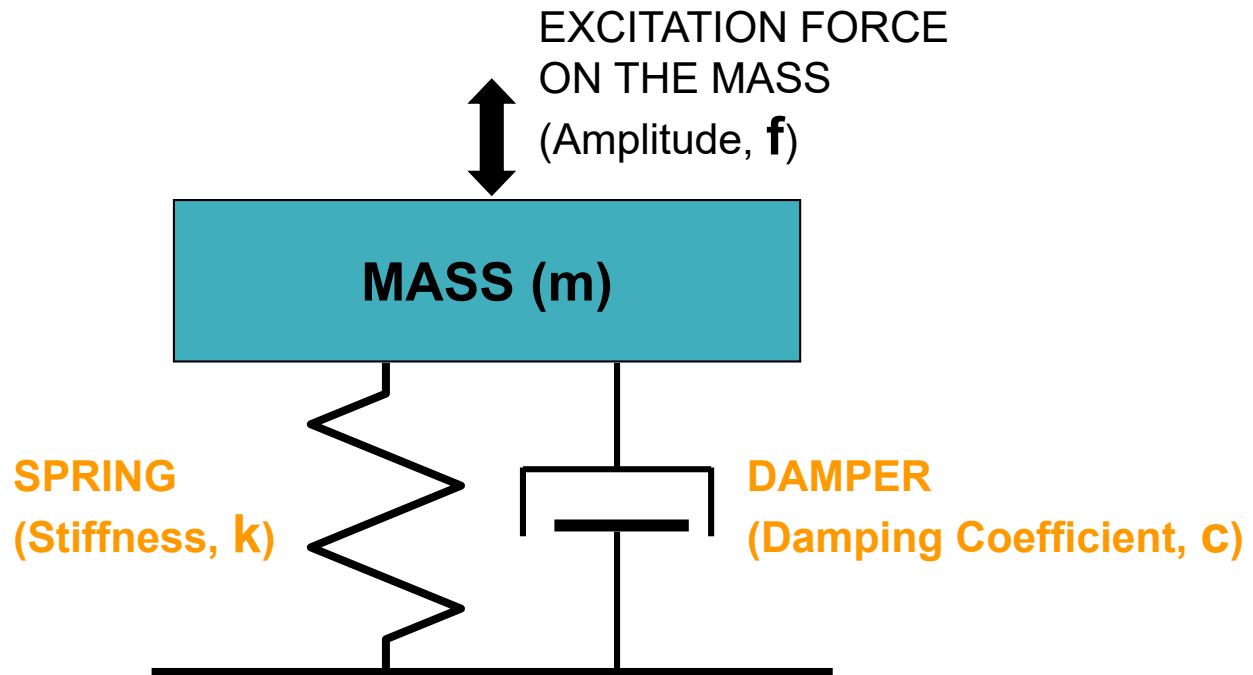
ARC4443 Acoustical, Electrical, and Illumination systems

Week 15-2

Vibration Isolation

VIBRATION BASICS

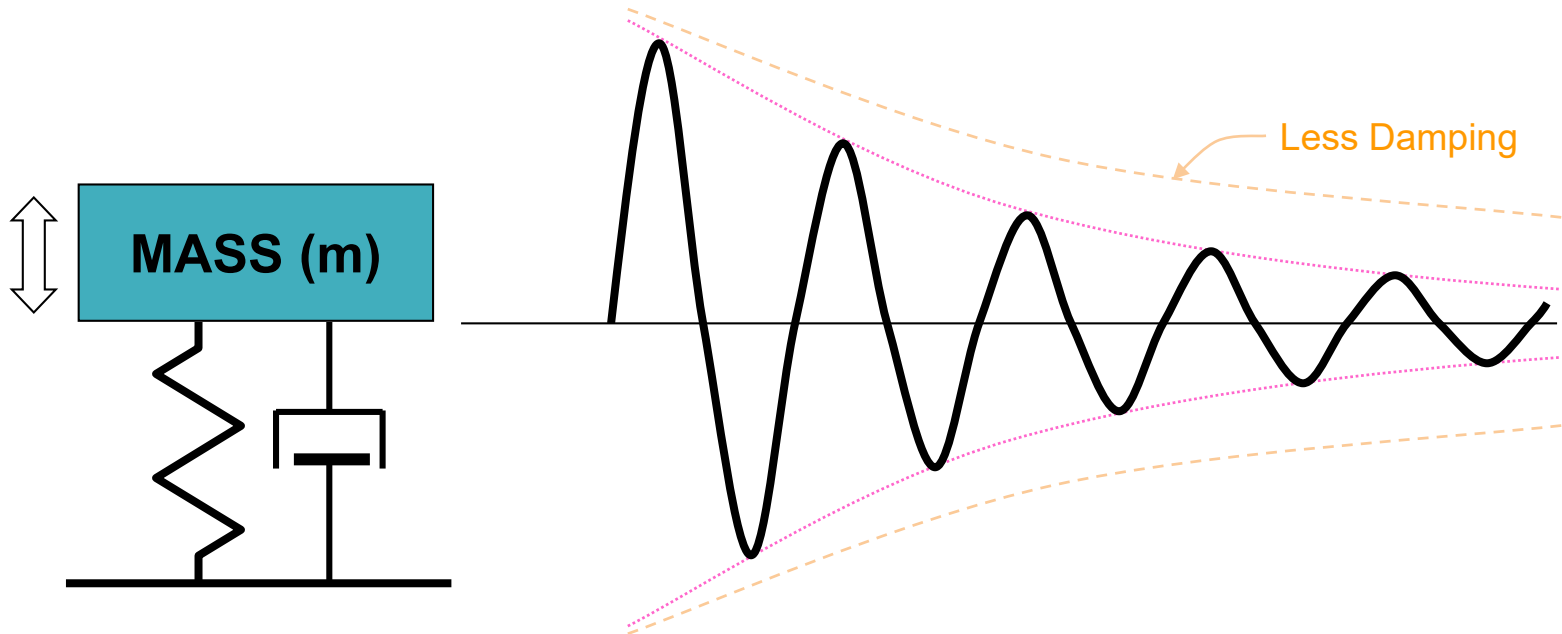
SIMPLE VIBRATION SYSTEM



The main elements of a vibration system are shown above. Most systems, however, are more complex and are represented by more than one of each element.

VIBRATION BASICS

DECAY OF A DAMPED VIBRATING SYSTEM AFTER IMPACT



When impacted, the mass will vibrate at its resonance. The level of vibration will decay and eventually stop. **The rate of the decay depends on the amount of damping.** A higher the decay rate indicates more damping in the vibrating system.

Natural & Forcing Frequency

Avoid resonance conditions!!

- Tacoma Narrow Bridge “Gallopig Gertie” – aeroacoustic flutter
- Broughton Suspension Bridge – failure from soldiers in step – “break step” to avoid
- Fox Theater, Detroit – balcony resonance, now balanced by dynamic damper
- Examples from BrightHubEngineering.com
 - Flue gas moving over second pass coils, the vortex shredding creates resonance of the gas columns - vibrations lead to failure of the tubes
 - Boiler safety valve failures from resonance of the steam column in the branch pipe on which the safety valve is mounted
 - Fatigue failure of thermowells (protective tubes for temp sensors) in high pressure steam piping due to flow induced resonance
 - Failures of pump shafts due to resonance at critical speeds and resonance due to frequencies from VFDs
 - Pump base plate resonance leading to piping vibration and failure
 - Failure of turbine shafts in Mohave power plant due to sub-synchronous resonance from series capacitors in the transmission lines

Fundamentals of Vibration Isolation

Sources of vibration

- Moving parts in equipment (rotation, reciprocation)
- Thermal processes (combustion)
- Aerodynamic (airflow in duct, fan interactions with air)

Unbalances in moving parts causes forces that move the equipment around

- Up & down, side to side, rocking motion

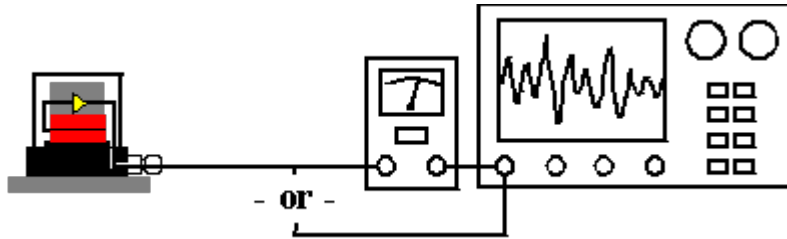
Vibration control is important because:

- Unbalances can decrease equipment life
- Vibration can become structure-borne noise with lightweight structures and all concrete structures especially susceptible

Vibration Measurements

- Instrumentation

- **Transducer/Accelerometer** – senses vibrational motion (piezoelectric)
 - Has a limited operable frequency range (just like microphones)



- **Preamplifier** – amplifies electrical signal
- **Analyzer**
 - Octaves, 1/3 octaves, 1/12 octaves
 - Narrowband (a.k.a. FFT)



Vibration Measurements

- Typical applications
 - Determination of source of vibration (if unknown)
 - Determination of structural resonances (to avoid)
- Comparison to equipment specs, building specs or general guidelines
- Comparison of vibration levels to expected machinery levels
- Comparison above & below the isolators to determine performance

Vibration Control

- *Vibration isolators* are used to reduce the amount of vibration that is transferred from the equipment to the structure
- Vibration reduction is especially important in
 - *Facilities that have vibration sensitive processes*
 - laboratories, operating rooms, magnification equipment, electronic circuit fabrication
 - *Facilities where excessive structure-borne noise can be radiated as airborne sound*
 - Lightweight construction

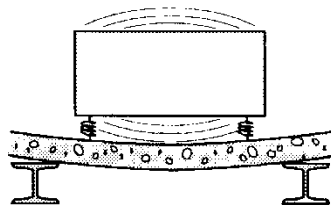
Vibration Control

- As often as possible, locate vibrating equipment on grade. The more rigid the floor, the easier it is to vibration isolate equipment.
- If located on an upper floor, the floor should have sufficient rigidity (i.e., small deflection under static load)
- Locate equipment on stiffest part of a suspended floor
In order of preference:
 1. over a column or load-bearing wall
 2. next to a column
 3. along a beam line between columns
 4. avoid in the middle of spans

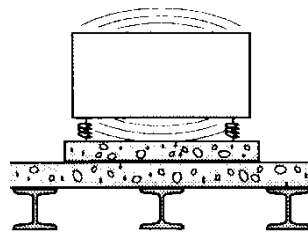
Vibration Control

Locate building equipment above columns or on grade when near noise or vibration sensitive spaces

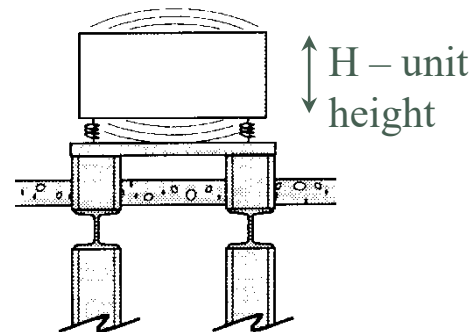
- **Poor:** high flexibility in mounting surface
- **Good:** stiffer mounting surface (more beams and/or concrete under unit) Note: Concrete pads also work to block airborne sound, particularly if extended out from the unit a distance of H from the unit perimeter.
- **Excellent:** on grade, above columns and/or on grillage (structural steel above roof supported from columns)



Poor



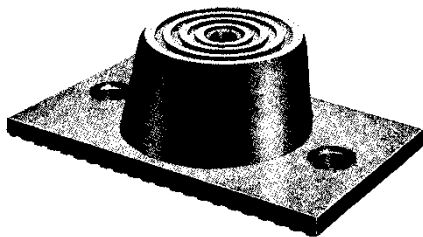
Good



Excellent

Isolator Types

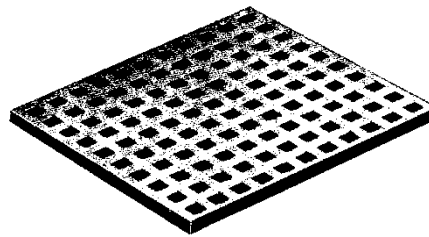
- **Neoprene isolators** typically have relatively low static deflections on the order of 0.10" to 0.45" (~10-20 Hz)
- High internal damping, so resonance is not typically an issue
- Don't overload pads: max 15% deflection of free height
- **Use bushings** to avoid metal-to-structure short circuit through pads



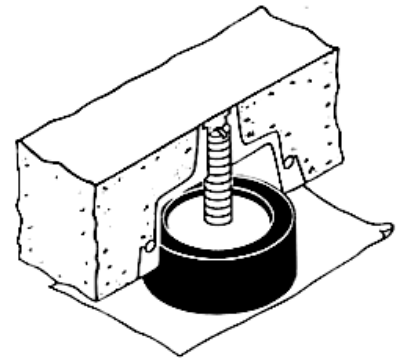
floor mounted
isolator



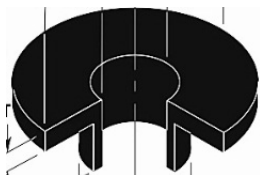
hanger



pad



jack-up floor mount



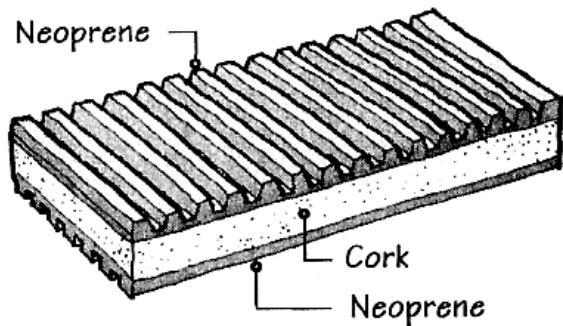
bushing



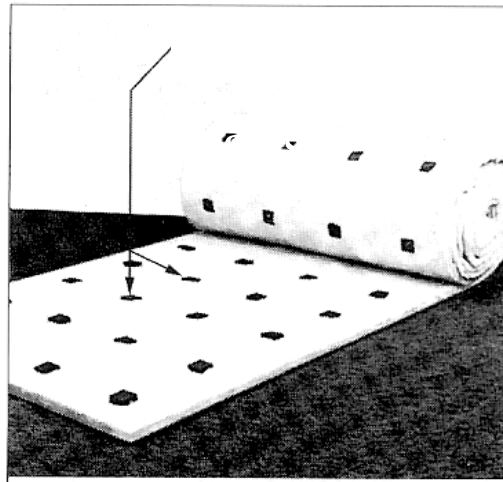
stacked pad

Isolator Types

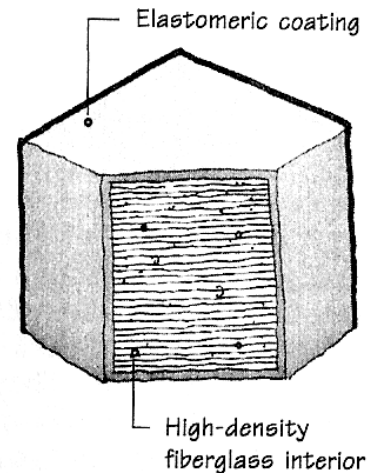
- Other pad isolator types: [cork](#), [high-density fiberglass](#), [foam](#)
- Cork is used in environments with temperature extremes because neoprene and rubber properties change with temperature
- Fiberglass is typically used to isolate floating floors and foundations



neoprene and cork pad



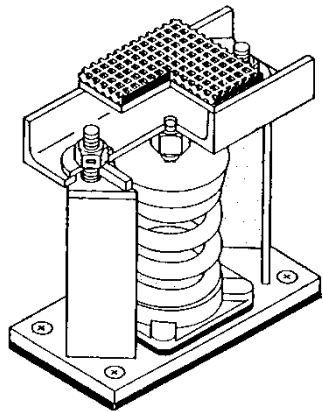
roll-out isolation material
for floating floors



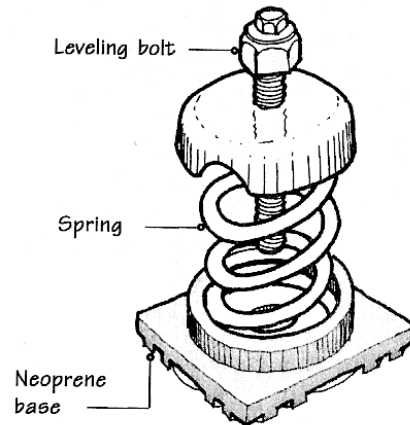
high-density
fiberglass
block

Isolator Types

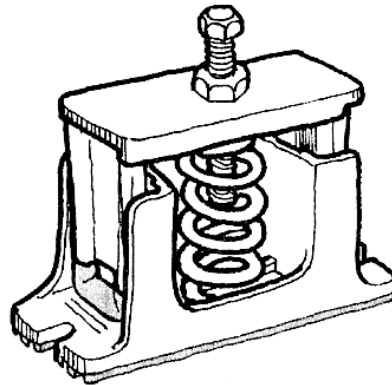
- Spring isolators have nominal deflections of 1" to 5" (~1.5-3 Hz)
- Can transmit high frequency through the spring coil (at its resonance), so neoprene is used in series with spring to isolate this high frequency



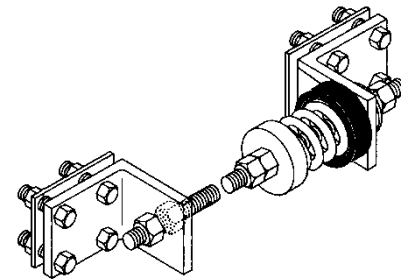
restrained



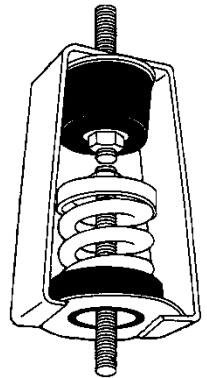
unhoused



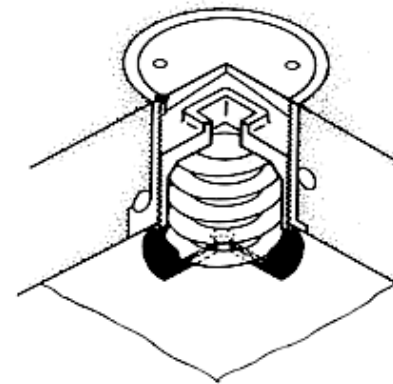
housed



thrust restraint



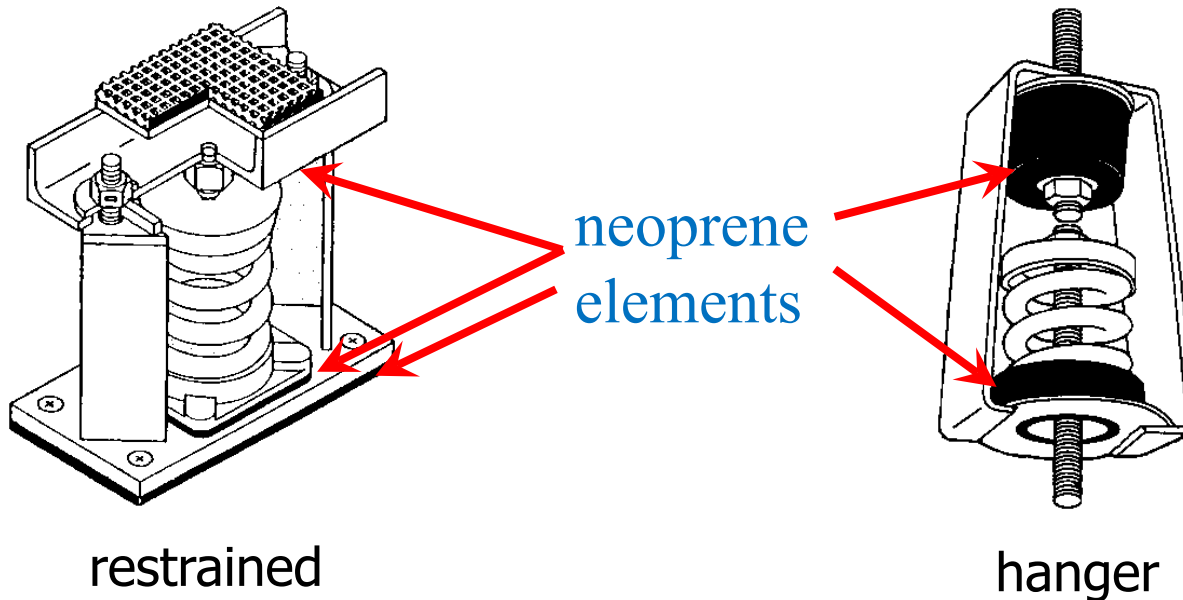
hanger



jack-up floor mount

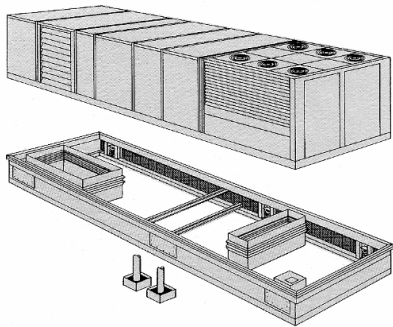
Isolator Types

- Spring and neoprene are used in series to prevent the spring surge frequency from transmitting energy
 - Surge frequency is the natural frequency of the coil
 - Can be an issue with screw compressors, inverters
 - Would typically occur above 200 Hz

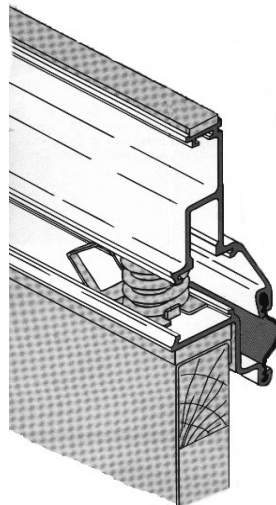


Isolator Types

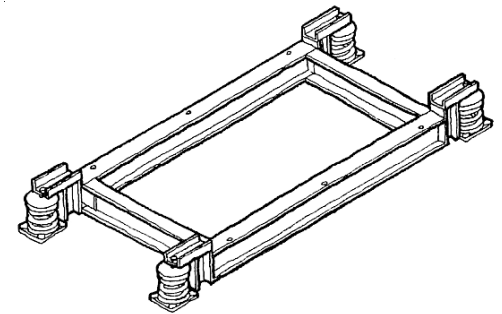
- **Bases:** framework to support equipment (**fans**)
- **Inertia base:** concrete filled structural steel base (**pumps**)
- **Curb mounted base or rail:** sit on top of roof curb
- **Spring isolation curb:** equipment curb with integral springs (**RTU**)



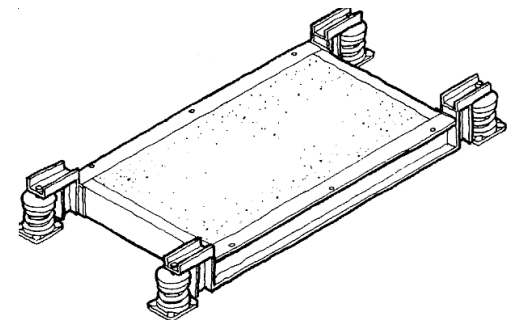
spring isolation curb



curb
mounted
base or
rail
(section)



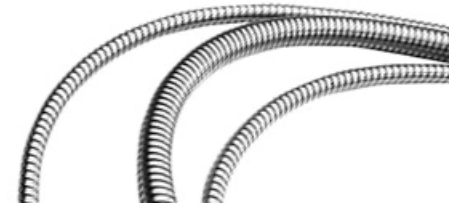
Structural Steel Base



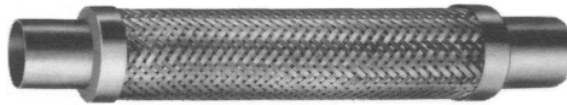
Inertia Base

Isolator Types

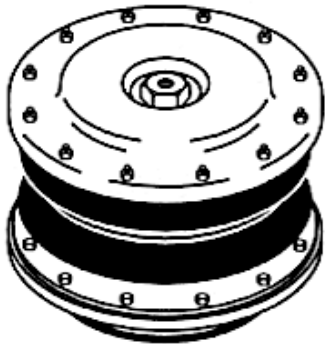
- **Air mounts** isolate to lowest $f_n \sim 1-3$ Hz
 - Costly, need constant air supply, only for critical applications
- **Isolation joint** in a building isolates a whole structure from the adjacent structure.
- All piping, duct and electrical connections to isolated equipment or vibration isolated structures must have **flexible connectors**



flexible conduit



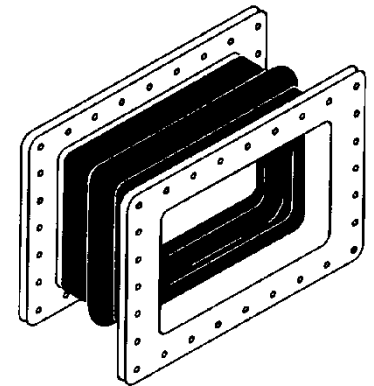
braided hose –



air mount



flexible pipe
connector
(double sphere)



flexible duct
connector

Isolator Types

Braided hose

- Not very flexible in axial direction
- Use perpendicular to the direction of vibration for greatest range of motion

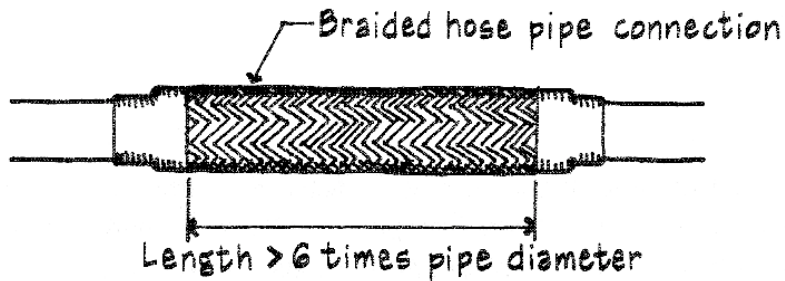


Table 48 Recommended Live Lengths^a of Flexible Rubber and Metal Hose

Nominal Diameter, in.	Length, ^b in.	Nominal Diameter, in.	Length, ^b in.
0.75	12	4	18
1	12	5	24
1.5	12	6	24
2	12	8	24
2.5	12	10	24
3	18	12	36

^aLive length is end-to-end length for integral flanged rubber hose and is end-to-end less total fitting length for all other types.

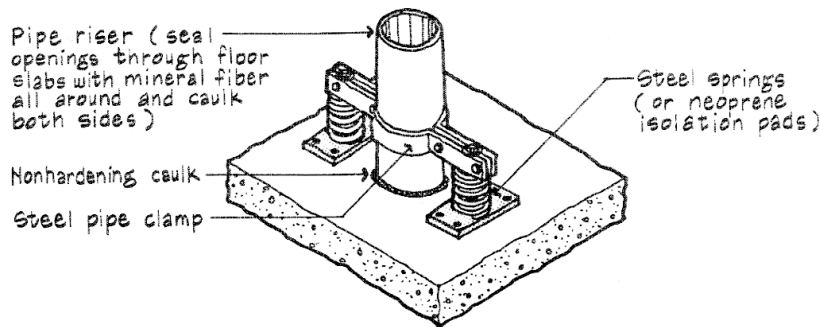
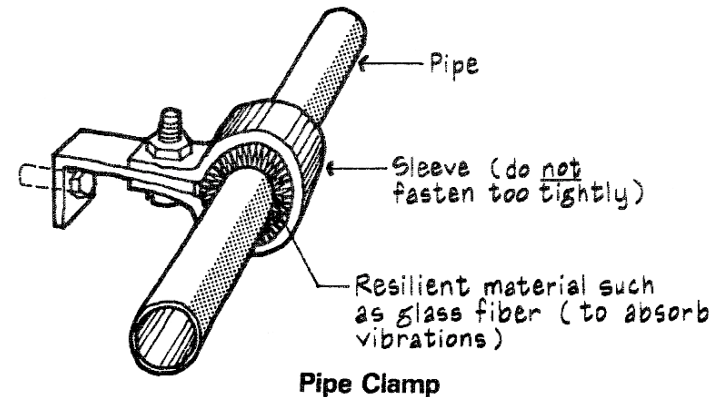
^bPer recommendations of Rubber Expansion Division, Fluid Sealing Association.

Isolator Types

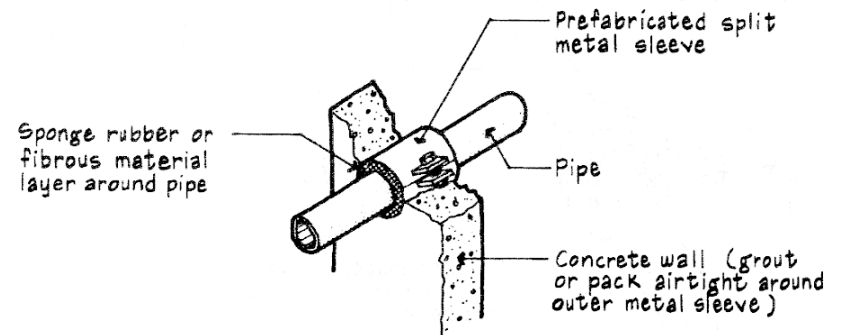
■ Piping isolation

- Where needed to minimize transfer of sound to the structure, use resilient materials between pipe and structure
- Do not allow pipe to touch the structure

■ Prevent water hammers



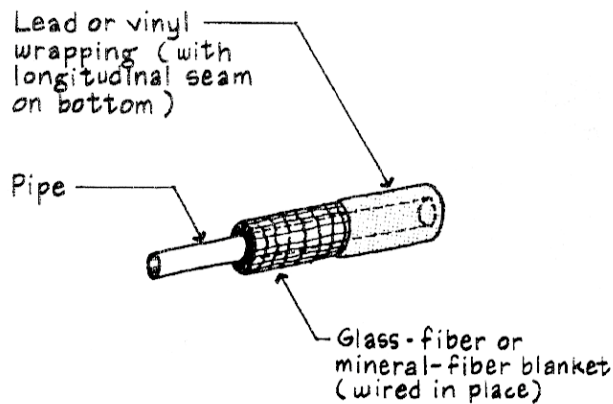
Floor Penetration



Wall Penetration

Isolator Types

- Use slower velocity flow from the table when piping has to be near acoustically sensitive spaces
- Alternately, lag the pipe with a sound absorbing material covered by a mass loaded vinyl wrap
- Use heavier walled pipe (i.e., cast iron instead of PVC)
 - Especially important for waste lines



Pipe Lagging Details

Pipe Inside Diameter (in)	Maximum Flow Velocity (ft/s)
1	3
2	4
3	6
4	7
6	8
8 and larger	10

Vibration Generating Building Equipment

- **Low frequency vibration** – spring & neoprene isolators
 - Generators, fans, AHU, pumps, air compressors, vacuum pumps, chillers, condensing units, dry coolers
- **Mid & high frequency vibration** - neoprene isolators
 - Transformers, theatrical dimming cabinets, valves (particularly steam)
- **Small equipment** – springs or neoprene
 - Heat pumps, fan powered terminal units
- **Cooling towers & boilers** – depends on type
- **Elevators**

Selection of Isolator Type and Deflection

Considerations

- Required isolator efficiency
- Mounting surface stiffness
 - Longer floor spans have higher deflection
- Equipment operating speed
- Equipment size/power

Specify isolator deflection

- Advise against letting contractor do selections for you
- Use ASHRAE isolation chart for standard construction, rule of thumb, or stricter criteria
- Incorporate into schedule or specifications

Selection of Isolator Type and Deflection

Internal vs. External vibration isolation

- Internal – only moving parts are isolated but not casing
 - **Advantages:**
 - Reduces load on isolators and consequently cost
 - Reduces vibration into housing
 - **Disadvantages:**
 - Not easily visible in the field to ensure proper operation
 - Isolator support point could be near the middle of a beam, which may provide inadequate stiffness for isolator
 - Short-circuiting of housed isolators by horizontal fan thrust can occur
 - Manufacturer's isolator type and deflection not based on specific building and may not provide enough vibration control
 - Does not reduce vibration in equipment housing caused by air movement
- Example: AHU fan base on spring isolators versus entire unit on spring isolation curb

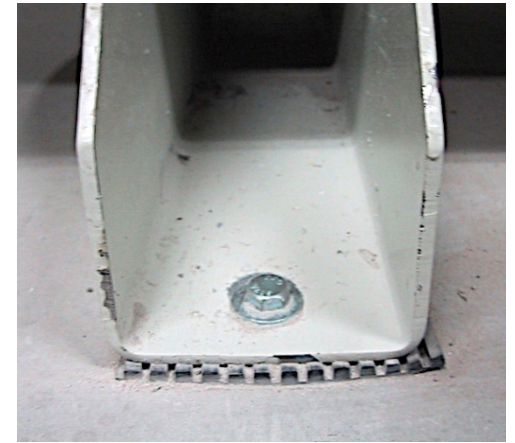
Isolator Short-Circuits

- Isolators are “**short-circuited**” or rendered nonfunctional when a solid connection occurs across the isolator
 - *Overloaded collapsed springs*



Isolator Short-Circuits

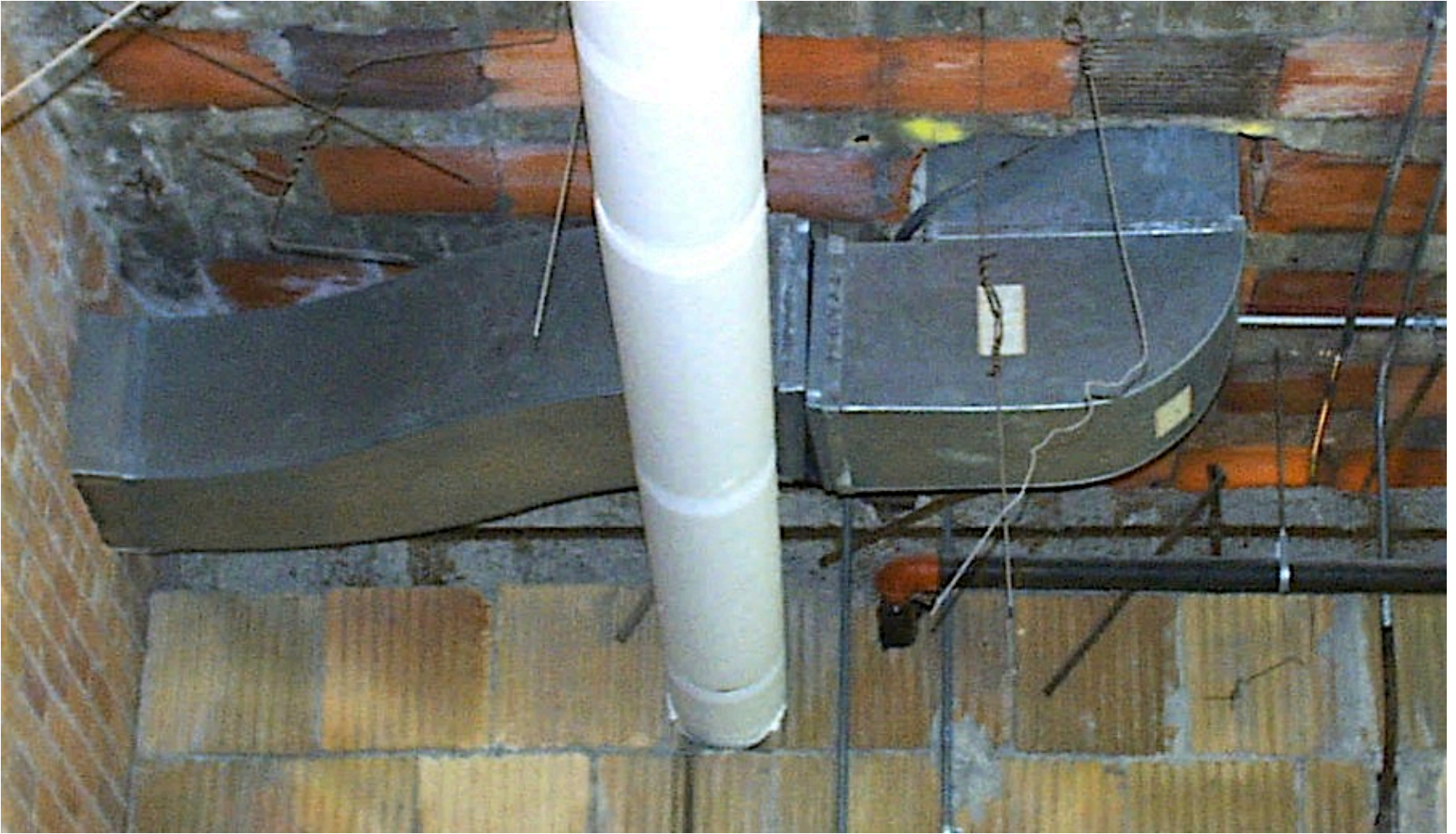
- Debris caught between isolated base and floor or housekeeping pad
- Equipment on isolator pad but bolted through base to floor without a neoprene grommet



Isolator Short-Circuits



Isolator Short-Circuits



Isolator Short-Circuits



Reference

- M. David Egan, Architectural Acoustics (Acton, MA: Copley Custom Textbooks, 2008) ISBN 1-58152-507-9
- Mechanical & Electrical Equipment for Buildings (MEEB), 12th Edition, Walter T. Grondzik and Alison G. Kwok, John Wiley and Sons, Inc., 2015, ISBN: 978-1-118-61690-4